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Machine learning based concept drift detection for predictive maintenance

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ABSTRACT

In this work we present a machine learning based approach for detecting drifting behavior – so-called concept drifts – in continuous data streams. The motivation for this contribution originates from the currently intensively investigated topic Predictive Maintenance (PdM), which refers to a proactive way of triggering servicing actions for industrial machinery. The aim of this maintenance strategy is to identify wear and tear, and consequent malfunctioning by analyzing condition monitoring data, recorded by sensor equipped machinery, in real-time. Recent developments in this area have shown potential to save time and material by preventing breakdowns and improving the overall predictability of industrial processes. However, due to the lack of high quality monitoring data and only little experience concerning the applicability of analysis methods, real-world implementations of Predictive Maintenance are still rare. Within this contribution, we present a method, to detect concept drift in data streams as potential indication for defective system behavior and depict initial tests on synthetic data sets. Further on, we present a real-world case study with industrial radial fans and discuss promising results gained from applying the detailed approach in this scope.

1. Introduction

With the increasing amount of data available from today's sensor monitored industrial machinery, expectations among operators, production managers and data analysts concerning potential valuable insights, are rising accordingly. One prominent approach originating from the ongoing digital transformation of industry is Predictive Maintenance (PdM). Its basic idea is to monitor and analyze a system in real-time in order to trigger maintenance proactively and prevent total breakdowns (Selcuk, 2017). While corrective maintenance implies to fix something the moment it breaks and preventive maintenance means to work on the basis of empirically defined maintenance schedules, predictive maintenance relies on a system's actual condition. For this purpose, machines are equipped with sensors and connected to a business intelligence unit in order to continuously monitor, learn and predict their behavior. As an ultimate goal of predictive maintenance, estimating a production machine's exact Remaining Useful Lifetime (RUL) is frequently mentioned (Saxena, Goebel, Simon, & Eklund, 2008). However, data sets which allow to make such ambitious lifetime predictions are difficult to gather. Since the reasons for defective behavior of industrial machineries are manifold, it would be necessary to carry out a large number of cost intensive *run-to-failure* experiments on entire fleets of machines, with no guarantee for valuable outcome. Hence, such real-world implementations of predictive maintenance are still rare.

In contrast to the envisaged lifetime prediction, our approach is to reveal the necessity of maintenance on a microscopic level. We aim to detect changing system behavior – so-called concept drifts (Tsymbal, 2004) – as an indication of beginning malfunctions. Therefore, we employ machine learning methods to model healthy systems on a component level in detail and identify deviations from the modeled state, when evaluating them on a continuous data stream.

In Section 2 an overview regarding related work in the field of machine learning based approaches for predictive maintenance and a distinction from the content of the present work is given. Subsequently, Section 3 details the algorithm developed in the scope of this contribution. Therefore, the algorithm pipeline and its components (e.g. data preprocessing techniques, machine learning algorithms, evaluation routines) are described. In Section 4 results from tests on synthetically generated data are summarized. Further on, we present results from a real-world case study concerning industrial radial fans in Section 5, for which the developed algorithm has been tested. Based on the performed experiments, we discuss the applicability of the approach, focussed on in this work. Finally, we conclude with a brief summery, as

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well as possible further directions in Section 6.

2. Related work and unmet needs

A fairly large number of recent contributions with reference to predictive maintenance is dedicated to software frameworks or reference architectures (Lee, Kao, & Yang, 2014; Li, Wang, & Wang, 2017; Sayed, Lohse, Sondberg-Jeppesen, & Madsen, 2015; Wang, Zhang, Duan, & Gao, 2017). While these mostly elaborate on the beneficial integration of recent technological developments such as Internet of Things, Cyber-Physical Systems, Digital Twins, Cloud Computing or Big Data Analytics, they do not necessarily detail the data analysis tasks. Although this work is also driven by a real-world application, not the technical setup and implementation, but the applied data analysis methods are the main concern.

In this context the term *predictive analytics* is frequently used, which basically refers to the adaption of statistical methods, such as machine learning algorithms for tasks such as predictive maintenance. Among the most frequently employed algorithms are: Linear Regression, Random Forest Regression (Mattes, Schöpka, Schellenberger, Scheibelhofer, & Leditzky, 2012), Boosted Trees (Cerqueira, Pinto, Sá, & Soares, 2016), Bayesian Networks (Mattes et al., 2012; Lee et al., 2014), Markov-Models (Cartella, Lemeire, Dimiccoli, & Sahli, 2015), Support Vector Machines (Widodo & Yang, 2007), or Artificial Neural Networks (Li et al., 2017; Kanawaday & Sane, 2017). A closer distinction between the approaches, based on what the actual prediction goal is, shall be made at this point: Some contributions aim for the prediction of remaining useful lifetime concerning the respectively monitored system (Cartella et al., 2015; Lee et al., 2014; Mattes et al., 2012). However, this requires large sets of data covering many different runs-to-failure (Saxena et al., 2008) for every system type in order to get reliable results. Others, including this contribution, pursue detecting and predicting systematic deviations from known healthy system states, which enables to trigger maintenance actions if a certain deviation threshold, defined by a domain expert, is transgressed (Li et al., 2017; Kanawaday & Sane, 2017). Systematic changes in data are well-known as concept drifts (Widmer, 1996), for which examples of research include weather forecasting, traffic monitoring, medical decision aiding, bankruptcy prediction, biometric authentication as well as the application in the focus of this work: industrial system monitoring. The recent surveys of Žliobaitė, Pechenizkiy, and Gama (2016) and Krawczyk, Minku, Gama, Stefanowski, and Woźniak (2017) provide a comprehensive overview for concept drift categorizations, application examples, as well as detection and learning techniques. Among the most popular concept drift detection methods developed over the years are: Drift Detection Method (DDM) (Gama, Medas, Castillo, & Rodrigues, 2004), Early Drift Detection Method (EDDM) (Baena-García et al., 2006), ECDD (Ross, Adams, Tasoulis, & Hand, 2012) and Linear Four Rates (LFR) (Wang & Abraham, 2015). These methods are based on statistical process control and have been designed for dealing with different kinds of classification tasks and various types of drifts. The approach, presented in this work shares their basic idea: If the estimation error of a preliminary built model increases, while being evaluated on a continuously updated set of data, up to a certain user-defined level, a concept drift is recognized.

This work differs from the stated methods as it concentrates on applying and adapting drift detection for a regression problem situated in a time-critical environment. A closely related example for regression model based concept drift detection is given in Winkler et al. (2015). Therein, data streams with changing behavior are attempted to be modeled with Symbolic Regression based on Genetic Programming in a sliding window fashion. Depending on the training success for each data partition, stable or drifting behavior can be determined with this method. However, since the detection is performed during the time consuming model training phase, this approach is not feasible for being applied on a continuously updated data stream with condition monitoring data. Therefore, in Zenisek et al. (2017), model training and test phases are split, which enables extensive search for accurate models in the time-uncritical *offline* phase and fast *online* evaluation when applying the model on streaming data. This work enhances the regression model based concept drift detection with a forecasting component and hence, poses a step towards concept drift prediction.

Similarly to this work, first experiences of the developed methodologies in the area of predictive maintenance are often gained using synthetic testbeds (Cartella et al., 2015; Kroll, Schaffranek, Schriegel, & Niggemann, 2014; Wang et al., 2017), before applying them to the field. Nevertheless, the topic has already been discussed and partly implemented on different levels for industry sectors, such as aircraft (Samaranayake & Kiridena, 2012), vehicle engines (Lee et al., 2014) and various manufacturing processes (Kanawaday & Sane, 2017; Mattes et al., 2012; Scheibelhofer, Gleispach, Hayderer, & Stadlober, 2016). However, the specific field of monitoring industrial radial fans has not been covered yet. These systems are, for example, installed for dedusting large production halls and hence, rank among crucial equipment for many manufacturing businesses which are absolutely necessary to be kept alive and ongoing. In the scope of Section 5 we elaborate on past and present research in this area and elaborate on our approach towards a predictive maintenance adaption for radial fans.

3. Machine learning based drift detection approach

The approach in this contribution relies on applying machine learning algorithms to model the state of a healthy industrial machinery, in which it is allegedly working “normal” and alert if this *concept* starts to drift towards “unknown” or potentially “defective”. Strategies for detecting drifting behavior in data streams, emerging from condition monitoring, can be divided in two categories.

- Classifying data streams as normal, drifting or defective based an a comparison with a large instance pool of formally under supervision categorized data sets.
- Predicting the next values in a data stream with a regression method and comparing these estimations with actually sensed values; an increasing prediction error might indicate an underlying concept drift.

In this work we use different machine learning methods following the second approach: regression model based concept drift detection and prediction.

Fig. 1 illustrates the generalized version of the data processing workflow from the case study dealt within this work, which consists of two phases – an *offline* model building phase (upper part) and an *online* model evaluating phase (lower part). In the *offline* phase a preprocessing pipeline, which incorporates tasks such as data consolidation, filtration, aggregation or transpositioning, is set up and initially fed with bulk data from the monitored system’s log files or database. In Section 3.1 we elaborate on basic preprocessing actions, while Section 5 details necessary steps for a real-world problem. The preprocessed data is subsequently used to train models in order to formalize normal machinery behavior. Therefore, we compiled a small set of prominent machine learning algorithms – Linear Regression (LR), Random Forest Regression (RF), Symbolic Regression based on Genetic Programming (SR), described in Section 3.2 – in order to develop two different model types: *state detection models* to verify system conditions and *time series forecasting models* to predict time series in a *n-step-ahead* fashion. Mainly, however, this study aims to shed light on the matter of how to apply the developed models for the task of concept drift detection, hence, the way they are evaluated on a constantly updated stream. After a brief introduction to data preprocessing and the applied machine learning methods, two algorithms are presented. One enables concept drift detection (see Section 3.3), the other is aiming for concept drift prediction (see Section 3.4). The entire approach has been implemented using the open source framework

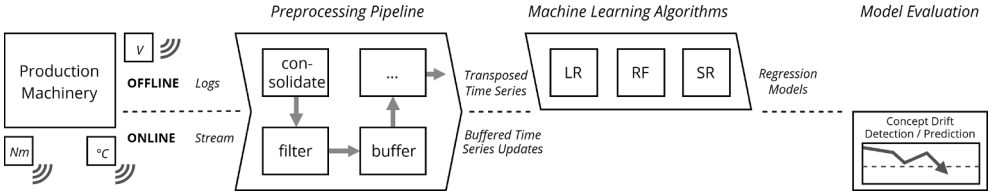


Fig. 1. From raw sensor data to concept drift detection.

HeuristicLab[1] (Wagner et al., 2014).

3.1. Preprocessing

The success of machine learning algorithms largely depends on quality and form of the data. Handling missing values, filtering outliers or correcting time-related irregularities are examples for problem specific data preparation tasks. For the case study, these preprocessing steps are described in Section 5. However, dealing with time series does also require some problem unrelated, generally applicable transformations, further described in this section. The objective of these is to shape the data so that arbitrary regression algorithms can be applied.

One prominent technique for time series preprocessing is decompositioning. Applying so-called STL – i.e. Seasonal-Trend decomposition based on Loess (Cleveland, Cleveland, McRae, & Terpenning, 1990) – splits a series into parts for season, cycle, trend and noise, which subsequently may be analyzed separately. Another technique is the spectral analysis of time series, which originates from the field of signal processing (Stein, 2016). Therein, the series is transferred to the frequency domain using Fourier transform, which decomposes a series' periodic (i.e. invariant) components into a combination of weighted sine and cosine functions, which can be used for further investigation.

However, in this work we focus on a simpler and hence, faster preprocessing method. New features are extracted by transposing the historically seen last n values in a series to a new input vector, which is subsequently provided to the machine learning algorithm. This is referred to as *backshifting* or *generating time lagged values*. For the experiments performed in this study the series have been backshifted between 5 and 50 times. Considering a sample of 1000 events and 5 input variables and a maximum lag of 50, preprocessing the data results with a matrix consisting of 1000 rows and 255 columns. With such matrices, machine learning algorithms are subsequently capable of developing time series regression models (see Eq. (1)). In order to enable the evaluation of these models also *online*, values have to be buffered in a First-In-First-Out (FIFO) fashion. In this work we applied backshifting to train the *time series forecaster models*, but not to train *state detection models* for identifying the current system condition (see details in Section 3.3).

$$x_t = f(x_{t-1}, x_{t-2}, x_{t-3}, \ldots x_{t-25}, \ldots) \quad (1)$$

In analogy, a similar lagged-value vector may be built by computing the numerical differences between the last n values. Accordingly, machine learning models, built with a differentiated input vector, forecast the next numerical change instead of the upcoming real value. Another, related possibility for feature extraction would be computing the moving average of the last values of a series. In preliminary work these time lagging enhancements, however, have been tested only with mediocre results (Ahmed, Atiya, Gayar, & El-Shishiny, 2010) and hence, all experiments in this work have been performed on solely backshifted series. We plan to investigate opportunities with decomposition techniques as well as alternative time lag variants in future work.

[1] http://dev.heuristiclab.com

3.2. Applied machine learning algorithms

For both used regression based model types, *state detectors* and *time series forecasters*, various machine learning algorithms may be applied, since the general approach is agnostic to underlying modeling techniques. In the scope of this study, we focus on a small set of prominent algorithms to support this statement. The selected algorithms create regression models based on prepared training data sets with the aim to describe one dependent target variable with one or more independent input variables. The following enumeration briefly summarizes the applied algorithms and refers the reader to literature for further details.

Linear Regression (LR) refers to a multivariate linear combination of regression coefficients (i.e. constants and weights of input variables). The coefficients are estimated by the generalized least square technique (Neter, Kutner, Nachtsheim, & Wasserman, 1996).

Random Forest Regression (RF) is performed using an ensemble of uncorrelated regression trees, which are randomly generated using bagging and boosting to fit the given data. The target estimation is performed by averaging the tree estimations. Initially this algorithm was developed for classification (Breiman, 2001).

Symbolic Regression (SR) refers to models in the form of a syntax tree consisting of arbitrary mathematical symbols (terminals: constants and variables, non-terminals: mathematical functions), which can be seamlessly translated to plain mathematical functions. For target estimation syntax trees are evaluated top-down. Syntax trees are developed using the stochastic genetic programming technique from the field of evolutionary algorithms (Koza, 1994). Within this work, we used the genetic programming variant with so-called *Offspring Selection* (Affenzeller, Winkler, Wagner, & Beham, 2009).

Since Linear Regression is deterministic and parameterless, there is no need to configure anything else than a data split for model training and test. For tuning the parameters of the stochastic algorithms Random Forest and Symbolic Regression a grid search has been performed, which is reported in the experiment section. This work aims to show the general applicability of regression model creating machine learning algorithms to lay the foundation of the concept drift detection approach, which is in the center of this work. This is pursued using a small set of three prominent algorithms. However, it does not focus on fine tuning, neither on providing a large-scale comparison of those algorithms. A more comprehensive comparison of machine learning methods with focus on their strengths and weaknesses for the task would exceed the scope of this work and hence, could be content of

```
MAXLAG = n

// offline
prepare data set // contains no drift
train state detector model

// online
push first MAXLAG-1 data updates into buffer
foreach further data update
    push update into buffer
    evaluate state model on buffer
    calculate mean prediction quality within a sliding window
    trigger alarm if prediction quality drops below user defined-threshold
    remove last element of buffer
```

Listing 1. Concept drift detection algorithm.

subsequent work. A broad comparison of machine learning algorithms applied for a different, but related regression task has already been given in Ahmed et al. (2010).

Linear Regression and Random Forest Regression represent very popular algorithms for regression problems in general and time series regression more specifically (Ahmed et al., 2010), which have already been applied to the field of predictive maintenance (Mattes et al., 2012). Both provide vital reference points for testing Symbolic Regression, which has not yet gained as much attention regarding predictive maintenance. One reason for closer investigating Symbolic Regression is the capability of modeling non-linearities and having a larger extrapolation potential of the mined functions compared to Random Forest Regression. Moreover, the mathematical form of Symbolic Regression models matches the “natural language” for describing physical processes in manufacturing industry. This might lead to models which are highly accurate and furthermore, interpretable for domain experts to gain new insights.

3.3. Concept drift detection

The concept drift detection algorithm is outlined in Listing 1. After problem-specific data preprocessing, the algorithm continues with training a regression model, which is later used as system state – i.e. system condition – detector. The modeling target should be a feature which is dependent on and hence, representative for the system’s health. The other variables are used as training input. Optionally, the variables, could be preprocessed by backshifting as described in Section 3.1 and additionally used as input. This way, the state detector would able to consider each series’ history, rather than just the most recent value, which might lead to more accurate results. However, in this work, the state detector models are created without backshifted features, since the model estimation quality was already sufficiently high. In order to enable a more comprehensive system state detection – i.e. to consider more than one target variable, which is representative for the system condition – also multi-objective models or variable interaction networks (Kronberger, Burlacu, Kommenda, Winkler, & Affenzeller, 2017) could be trained. However, in this work we concentrate on the application of single-objective regression models, for which each of the described machine learning algorithms may be used to train.

The online phase of the algorithm is designed to buffer data value by value, until the number of used time lags -1 is matched. Each subsequently received value updates the FIFO-buffer and furthermore, triggers the evaluation of the created regression model, so that the defined state-representative feature is estimated. In order to detect concept drifts, the relation between these model estimations and actual sensor values is constantly monitored. If the model estimation error increases over the course of time, models do not fit the current data anymore, which might indicate changing system behavior. To prevent the evaluation from being prone to short-term anomalies and mine more general trends, the model estimation error is averaged within the boundaries of a sliding window. The size parameter of the sliding window allows to control the number of consecutive sensor values, i.e. the specific ”period of time” for which the model error is averaged, and hence, the sensitivity of the detection algorithm. Eventually, concept drift is detected, if the estimation error transcends a certain threshold, defined by an application domain expert. Hence, setting such a threshold is considered as problem-dependent, as it also implies to handle the tradeoff between drift detection reactivity and drift detection confidence. In 2 an exemplary threshold is illustrated, although this work does not aim to evaluate the algorithm’s performance based on such thresholds.

Experiments in the scope of our most recent projects revealed promising results using the described methodology and motivated for enhancing the approach in this work towards drift prediction (see Section 3.4).

3.4. Concept drift prediction

The outlined algorithm in Listing 2 enhances the concept drift detection algorithm with a forecasting component. As well as for the latter detection algorithm, first a state detector model, based on a system state representative feature, is trained. After preparing a second data set by generating backshifted values, for each input variable a *1-step-ahead* time series forecaster is created. Therefore, the selected machine learning algorithm trains regression models, using all the lagged values of the target variable and the first lagged value (i.e. the most recent value) of each other variable as input. This guarantees that a series’ internal as well as most recent external influences can be modeled. While the training data set for the state detector should be representative for a healthy system state, the data for modeling forecasters must incorporate drifting behavior. Only, if the models are trained on both healthy and drifting behavior representative data, they are able to forecast the present behavior when applied for concept drift prediction.

The online phase of this algorithm differs from Listing 1 by the utilization and the so-called *rolling horizon evaluation* of the forecasters. Therefore, the time series regression models are evaluated sequentially for m times. Each time, the resulting vector of estimations is used to update a buffer, such that beginning with the second evaluation round, the models use their own estimations to forecast. Since the successive evaluation of the state model is not performed on current, i.e. actually recorded data anymore, but on the forecasted streams instead, concept drifts may theoretically be *predicted*. Therefore however, the uncertainty of the underlying time series forecasts is crucial.

```
MAXLAG = n
HORIZON = m

// offline
prepare data set // contains no drift
train state model

prepare data set // must contain drift(s)
foreach input variable
  train 1-step-ahead forecasting model

// online
push first MAXLAG-1 data updates into buffer1
foreach further data update
  push update into buffer1
  copy buffer1 to buffer2
  for 1..HORIZON
    foreach input variable
      evaluate respective forecaster model on buffer2
      push prediction into buffer2

  evaluate state model on buffer2
  calculate mean prediction quality within a sliding window
  if prediction quality drops below user defined-threshold, trigger alarm
  clear buffer2, remove last element of buffer1
```

Listing 2. Concept drift prediction algorithm.

4. Synthetic data based experiments

Providing a reasonable testbed for concept drift detection on time series, which are similar to those from industrial condition monitoring, is a non-trivial task. There is only little real-world benchmarking data available, which was authorized for publication. In order to provide a realistic test environment, we follow a model based data synthetization approach.

4.1. Time series regression model based data synthetization

The basic idea of this data synthetization approach is to reuse mathematical time series regression models – e.g. results from applying the Box-Jenkins method (Box, Jenkins, Reinsel, & Ljung, 2015) or computing Fourier series by harmonic analysis (Stein, 2016) on real-world time series – in order to produce new synthetic time series with the characteristics of the original data. In the following, the complete synthetization specification is provided. First, we assume a sinusoid formula (2), representing a production machine's periodic behavior, similar to what has been obtained from a real-world data set. The function output x_1 subsequently represents the target variable of the *state detector* model. Based on Eq. (2), we derive several (more or less) dependent series, using $x_{1_{t-1}}$ as weighted additive term in other time series models, such as autoregressive ones (6), or other periodic functions (3). These derived series subsequently serve as input for the *state detector* model. As per Section 3.4, a *forecaster model* is trained for each of these series. In the following formulas t refers to the past time steps, implemented as integer counter variable which starts at 1 and increases by 1 for each generated value of a series. Moreover, ϵ represents an additive term, sampled from a Gaussian distribution $\epsilon \sim N(0, 0.02)$ which is used to simulate signal noise and thus, impede the eventual modeling task.

$$x_{1_t} = 1 + \epsilon_t + 0.5 * (sin(t/5) + sin(2 * t/5)/2 + sin(5 * t/5)/7) \tag{2}$$

$$x_{2_t} = \epsilon_t + x_{1_{t-1}} * 0.4 + cos(3.5 * t/5)/3 \tag{3}$$

Usually wear and tear are conceived as long-term trend, while damage or fatigue failure represent abrupt changes. To simulate different damage propagation patterns, exponential and logistic functions have been found to suit well for describing a machine's run-to-failure. Schönmann, Dengler, Intra, Reinhart, and Lohmann (2017) illustrate a macroscopic model with a sigmoid function for describing a *burn-in phase* at the beginning of a system lifecycle and increasing wear at the end of it. Similarly, Saxena et al. (2008) outline several mathematical models with exponential characteristics for simulating damage propagation and noise. Also linear increase of wear might be considered, e.g. for friction linings, which are stressed in a constant fashion. Based on these considerations, we defined a sigmoid function to simulate realistic gradual increase of wear (4). The resulting *deterioration series* will not be provided for training nor evaluating any regression model. It is utilized to influence the other sampling functions to some extent by changing the weight of sub-terms, as depicted in Eqs. (5) and (7). Changing the behavior of the input series, without adapting the target function for x_{1_t}, causes a change of the dependency between inputs to target. Hence, a *hidden* concept drift is introduced, which is not recognizeable from visually inspecting the data series.

$$d_t = 1/(1 + \exp^{-0.05 * (c-300)}) \tag{4}$$

$$x_{2_t} = \epsilon_t + x_{1_{t-1}} * (0.4 + 0.2 * d_{t-1}) + cos(3.5 * t/5)/3 \tag{5}$$

The sampling functions Eqs. (6)–(8) complete the set of input series. With Eq. (9) a second wear-simulating function is provided, which is used to generate the data streams for the final concept drift detection and prediction tests. This way all final tests are performed solely on unseen data.

$$x_{3_t} = 0.5 + \epsilon_t + \sum_{i=1}^{p} P_i y_{t-i} + \sum_{j=1}^{x_1} X_{1_j} y_{t-j} \tag{6}$$

$$P = [0.7, -0.2, 0.05], X_1 = [0.02, 0.005]$$

$$x_4 = \epsilon_t + x_{2_{t-1}}/x_{1_{t-1}} * (0.3 - 0.1 * d_{t-1}) + cos(0.2 * x_{2_{t-1}} * 0.2 * x_{3_{t-1}})/2 \tag{7}$$

$$x_5 = cos(0.6 * x_{2_{t-1}}) + 0.1 * x_{3_{t-1}} \tag{8}$$

$$d_{2_t} = 1/(1 + \exp^{-0.05 * (c-250)}) \tag{9}$$

The described approach may be summarized as a reversed and enhanced version of modeling Auto Regressive Moving Average (ARMA)

Table 1
Parameter grid search boundaries for *state detector* and *forecaster* models. Each modeling experiment has been performed with 10 repetitions.

Algorithm	Parameter	Value
RF	Number of Trees	25, 50, **100**, 500
	R	0.2, 0.3, 0.4, 0.5, 0.6, **0.7**, 0.8
	M	0.2, 0.3, 0.4, 0.5, **0.6**, 0.7, 0.8
SR	Tree Length	25, **35**, 50
	Population Size	50, **100**, 250
(fixed)	Type	Offspring Selection Genetic Algorithm
	Max. Generations	1000
	Function set	Unary functions (sin, exp, log)
		Binary functions (+,−,×,÷)
	Terminal set	*constant*, *weight·variable*
	Selection	Gender specific selection (proportional and random)
	Fitness function	Pearson's R^2 correlation with the target
	Crossover	Rate = 100% Subtree crossover
	Mutation	Rate = 25% Change symbol, single-point, remove branch, replace branch

models with exogenous parts (ARMAX) (Bauer, 2009) and trigonometric functions for sampling purposes. For a comprehensive description of related synthetization methodologies, the reader is referred to Winkler et al. (2015) and Zenisek, Wolfartsberger, Sievi, and Affenzeller (2018).

4.2. Experiments and results

The subsequently described experiments have been performed on a synthetic data set following the specifications in Section 4.1. For training data we generated series with 1000 events, while for test data – used for performing concept drift detection and prediction tests – 500 events were sampled.

4.2.1. Algorithms configuration

For both, building *state detection models* as well as *time series forecasters*, a grid search for suitable algorithm parameters has been performed. The search only applies for Random Forest and Symbolic Regression, since Linear Regression is deterministic and does not need parameterization. The boundaries of the search are provided in Table 1. The picked parameter configurations are marked bold if applied for detection models and underlined if applied for forecasters. The reader is reminded that this work does not focus on extensive algorithm tuning, but instead aims to test the presented algorithm-agnostic concepts.

4.2.2. State detector training

In Table 2 results from training the *state detection models* are provided. For each algorithm configuration the Average Relative Error (ARE), the Normalized Mean Squared Error (NMSE) and the squared Pearson correlation (PR^2) of the best model are listed. The training of the models was performed on a shuffled data set, representing a *healthy* state, with a training-test split (67%/33%). The most accurate detectors on the training partition have been created using Random Forest Regression (RF). However, the models built with Symbolic Regression

Table 2
Model estimation qualities of the *state detector* for the simulated data set.

	ARE		NMSE		PR^2	
	Train	Test	Train	Test	Train	Test
Linear Regression (LR)	0.179	0.188	0.199	0.184	0.800	0.816
Random Forest Regression (RF)	**0.063**	0.174	**0.030**	0.208	**0.974**	0.793
Symbolic Regression (SR)	0.115	**0.138**	0.108	**0.148**	0.891	**0.851**

Table 3
Model estimation quality PR^2 of input variable *forecasters* on the test partition of the simulated data set.

Lag	10			25			50		
	LR	RF	SR	LR	RF	SR	LR	RF	SR
x2	0.972	0.935	0.964	**0.989**	0.966	0.977	0.992	0.977	0.983
x3	**0.654**	0.603	0.645	0.623	0.584	0.610	0.646	0.603	0.564
x4	0.819	0.883	0.790	0.822	0.858	0.850	0.882	**0.892**	**0.892**
x5	0.942	0.904	0.914	0.956	0.949	0.921	**0.975**	0.966	0.960

(SR) outperform all others on the test partition. While RF models seem to *overfit*, the SR models perform better on "unseen" data. The Linear Regression (LR) models do not overfit, but they cannot compete with SR. However, all models represent very accurate state detectors and perform on a high level.

4.2.3. Forecaster training

The training of *input variable forecasters* was performed on a preprocessed (i.e. backshifted) and subsequently shuffled data set with a training-test split (67%/33%). A brief collection of performance results from the test partition with the measured PR^2, is given in Table 3. One can observe that also these time series regression models perform on a high level. Only the forecasters for $x3$ trail behind, which is presumably caused by the high level of noise in the series' sampling process. The best models have been accomplished with Linear Regression and a maximum of 50 backshifted values per series (cf. *Lag*).

4.2.4. Concept drift detection and prediction

Examples for the subsequently performed tests of the developed evaluation routines are illustrated in Fig. 2 regarding *concept drift detection* and in Fig. 3 regarding *concept drift prediction*. For this exemplary evaluation, we selected a Symbolic Regression model as *state detector* and Linear Regression models as *time series forecasters*, following the best stated test results. Furthermore, the models use a maximum of 10 lags, since the result qualities are comparable with a higher number of lags and the execution time decreases considerably with less lags, which is important for real-time applicability. The depicted series represent the quality measures of the evaluated *state detector models*, averaged within a sliding window with *size* = 100. One can easily observe from both figures that it was possible to *detect* and *predict* the synthetically introduced concept drift, since the progression of the computed quality measures strongly correlates with the deterioration waveform *DET*, available from the data sampling process: While the deterioration *DET* increases, all model error measures increase accordingly and the model-fitness indicating PR^2 decreases. Hence, as soon as a decline in quality measures becomes present, the model fails to describe the system behaviour and a concept drift can be assumed. The Mean Absolute Error (MAE), Mean Squared Error (MSE) and Root Mean Square Error (RMSE) are additionally listed, as a deterioration may not manifests in all quality measures and the proposed algorithm intends to be agonstic to the underlying quality measure, which could be shown.

In Fig. 2 we added an exemplary detection threshold for demonstration purposes. Such thresholds might be set by domain experts for real-world applications according to the problem data and the aimed detection sensitivity. However, for this work, it was not the goal to specify a certain point of time in order to be able to state "concept drift happened". Instead, we aimed to measure the quality of the developed drift detection approach. Since we knew the actual deterioration waveform from the artificial data sampling process, we were able to compare it with the model quality progression and calculate their correlation coefficient (see Fig. 4 and Table 2). This way, a temporally accurate assessment of the developed algorithms' performance was possible.

The visual comparison of Figs. 2 and 3 furthermore, indicates that

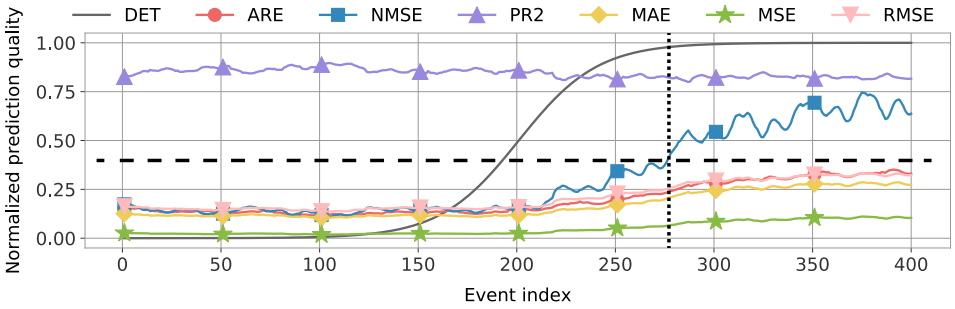


Fig. 2. Known deterioration DET vs. detection qualities (i.e. no forecasting). The horizontal dashed line refers to an exemplary NMSE-threshold, which is to be set by a domain expert with respect to the problem and the aimed detection sensitivity. In this (artificial) case, the threshold has been set to 0.4, which is approx. transgressed at event index 275, visualized by the vertical dotted line.

the prediction algorithm is able to capture the drift earlier as the detection algorithm, which depicts the beneficial working of the underlying time series forecasting models. However, this temporal advance comes with the price of a lower level of model estimation quality and stability (cf. generally higher error level and uneasier curves in Fig. 3). The following evaluation aims to shed light on the effect of increasing the prediction horizon.

In Table 4 a result collection, containing derived performance measures concerning the developed drift detection and prediction routines, is provided. In Fig. 4 an outline of the listed results is visualized and now discussed in greater detail: The shown performance measures represent the correlation coefficient, calculated between the known deterioration value on one hand and the quality measures from the evaluation of the state detector model on the other. The exact deterioration value is known at any time, since it is synthetically introduced with the provided specification (see Section 4.1). The results show that both routines, *concept drift detection* (i.e. $horizon = 0$) as well as *concept drift prediction* (i.e. $horizon > 0$), applied on the synthetic data set, are highly successful. We observed that (for a certain time frame) an increasing forecasting horizon up to a certain level (cf. $horizon < 25$), results in an increasing correlation. Within this time frame, the correlation is even higher than the value gained from concept drift detection (cf. $horizon = 0$), hence a prediction of concept drifts is possible. In the scope of a real-world implementation, this would mean that optimal points for maintenance actions could be determined and (to some extent) predicted, in real-time. However, with increasing prediction horizon the detection quality decays rapidly, no matter the monitored quality measure. Several reasons may be responsible for the limitation of the prediction horizon: On reason is the forecasting uncertainty, which increases together with an increasing rolling prediction horizon, since the trained 1-step-ahead forecasters are quite accurate but not perfect. Moreover, it is likely that with a too large prediction horizon, the effect of a drift is simply not yet present in the data of the specific frame, with which the algorithm was fed, which makes it impossible to predict the drift. As well as defining the optimal sliding window size, or determining a proper model error threshold, finding a feasible prediction horiozon is difficult and remains an application-dependent challenge when using the proposed methods.

The described methods represent a powerful and yet flexible framework for dealing with concept drift in data streams, as proved in real-world use cases (see Section 5).

5. Case study: radial fans

After testing the machine learning based approach on synthetically generated time series, verification on real-world data is the next logical step. Therefore, an experiment with real-world machinery was conducted. Scheuch GmbH, an industry partner, provided a new radial fan, commonly found in industrial applications (see Fig. 5). This radial fan was used for a run-to-failure, generating data for this case study.

5.1. Related work

Crucial for any health state modeling or health state prediction is the acquisition of relevant sensor signals. Generally, these signals are analyzed and the current state is derived from this recorded data. As knowledge of health state specific characteristics is gained, deviations can be detected. Because the origin and manifestation of faults are manifold, relevant literature focuses on different signals and different methodologies. As a starting point, some basic literature in the field of fault diagnosis and health monitoring shall be mentioned: (Adams, 2000; Gan, Wang, & Zhu, 2015; Goyal & Pabla, 2016; Rai & Upadhyay, 2016; Scheffer & Girdhar, 2004; Wang, Xiang, Markert, & Liang, 2016b). In the context of rotating machinery, vibration analysis is a frequently used method to detect faults (Jung, Zhang, & Winslett, 2017; Lv & Qiao, 2017; Yang, Dong, Peng, Zhang, & Meng, 2015; Rusiński, Moczko, Odyjas, & Pietrusiak, 2014; Renwick & Babson, 1985; Wang, He, & Kong, 2013). By applying different methodologies on the vibration data, such as Fourier transform, wavelet analysis or the use of symmetrized dot patterns (Wang, Liu, & Xu, 2016a; Xu, Liu, Zhu, & Wang, 2016), detection of a healthy or damaged state can be achieved. Other possible ways to detect an defective state may utilize the acoustic signals of (radial) fans by modeling the normal acoustic behavior and detect deviations (Abid et al., 2012; Dogan, Eisenmenger, Ochmann, & Frank, 2018; Khelladi, Kouidri, Bakir, & Rey, 2008; Sanjose & Moreau, 2017). Other research considered atmospheric pressure and volumetric flow (Xu, Wang, Liu, Li, & Wang, 2013; Velarde-Suárez, Ballesteros-Tajadura, & Hurtado-Cruz, 2006). Furthermore, some literature focuses on different origins or patterns of damage, such as crack detection (Hu & Li, 2015; Sun et al., 2017; Yu, Fu, Gao, Zheng, & Xu, 2018),

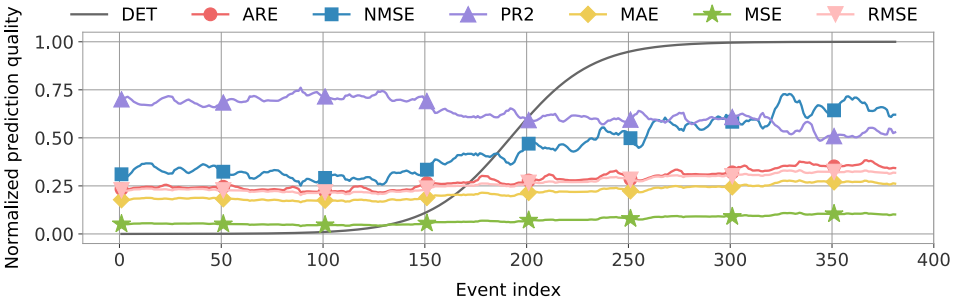


Fig. 3. Known deterioration *DET* vs. prediction qualities with forecasting $horizon = 25$.

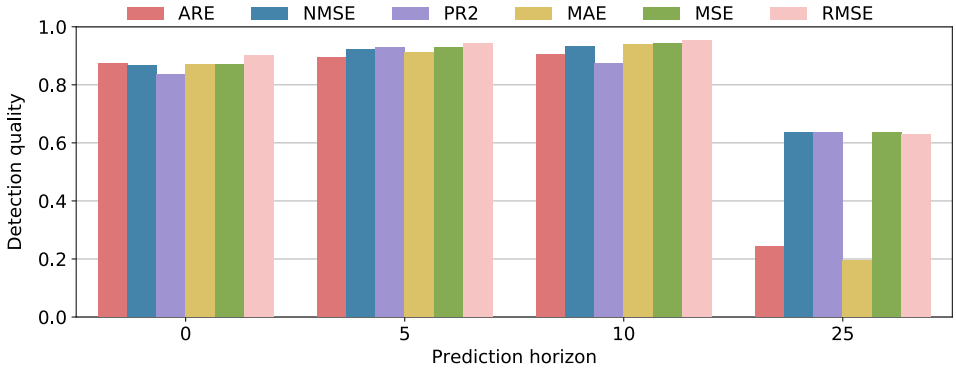


Fig. 4. Known deterioration vs. model prediction qualities.

Table 4
Known deterioration vs. model predictions qualities.

Hor.	ARE	NMSE	PR2	MAE	MSE	RMSE
0	0.872	0.865	−0.835	0.870	0.871	0.901
5	0.895	0.921	−0.927	0.911	0.927	0.943
10	0.905	0.931	−0.874	0.939	0.944	0.952
25	0.244	0.636	−0.636	0.195	0.636	0.628
50	0.561	0.863	−0.832	0.907	0.885	0.889
100	−0.714	−0.560	0.659	−0.678	−0.639	−0.639



Fig. 5. Radial fan used for data generation.

foundation damage (Eskandari, Nik, & Pakzad, 2016; Eskandari-naddaf, Gharouni-nik, & Pakzad, 2018) and damaged bearings (Al-Bugharbee & Trendafilova, 2016; Ding & He, 2016; Janssens et al., 2016; Miao, Zhao, Lin, & Lei, 2017; Orhan, Aktürk, & Çelik, 2006; Satishkumar & Sugumaran, 2017; Singh, Darpe, & Singh, 2017; Zhang, Zuo, & Bai, 2013). By applying multivariate regression methods, we can take multiple, different sensor signals into account and are not limited by vibration, acoustics, pressure or volumetric flow alone.

5.2. Setup

In preliminary studies, two impellers of which one was brand new and one artificially damaged, were used for a first approach on modeling the health state of a radial fan (Holzinger et al., 2018). The excessive degradation of an impeller is one of the most common failures of radial fans. Since a replacement of the impeller may only be necessary after prolonged periods of use, depending on the environment and usage, an alternative for faster degradation and eventual run-to-failures is advisable. For this purpose, a new experimental setup in a laboratory was arranged. This setup included an additional closed loop circulating abrasive material, namely silicon carbide, to simulate long-term abrasive stress in a shorter time frame. An additional filter was mounted to separate and extrude fine particles from the cycle, whereas coarse grained particles remained in the cycle. This was absolutely necessary since the abrasive material poses a health hazard and may contaminate the surrounding environment. The radial fan was powered by an electrical motor (37 kW) and a frequency converter allowed the adjustment of the rotational speed between 0 and 2940 rpm. A Venturi tube was installed, enabling the measurement of differential atmospheric pressure and therefore the volumetric flow. An additional inlet, specially designed for this setup, allowed a controlled release of abrasive material into the closed loop. This design minimized pressure loss and therefore mitigated a possible negative impact on the test setup. For this experimental setup, several new impellers were provided in order to be mounted and used for additional runs until a predefined grade of wear is achieved. This progression of wear was monitored via regular visual inspection and analysis of the sensor data (e.g. maximum vibration threshold of 11 mm/s reached), utilizing domain knowledge and the experience of an expert. As soon as either safety issues occurred or the setup showed signs of excessive wear, usually triggering a maintenance in real-world applications, the impeller was replaced. For this work, one complete run-to-failure was carried out and the whole process will be repeated with the remaining impellers for future tests.

After some initial test runs and experimentation, with the goal to approximate a feasible saturation of abrasive material in the airflow, the chosen configuration wears an impeller in about 160 h. Fig. 6 and Fig. 7 illustrate the impeller used for the run-to-failure in it's healthy and damaged state, respectively before and after the test run. In the interim stages of wear, caking could be observed until, under the constant influence of sandblasting, the worn impeller has finally breached and a replacement was necessary.

To gather data of the gradual degradation of an impeller, several different types of sensors were mounted on the radial fan (see Table 5). These sensors recorded the data used for drift detection. The different types of sensors were chosen based on previous work in the field of radial fans and general rotating machinery, detecting errors (see Section 5.1).

5.3. Data preparation

To prepare the data properly, several preprocessing steps were necessary. First of all, the complete run-to-failure produced about 250 GB of raw data, therefore a reduction of data was imperative. The vibration data was sampled with 1000 Hz, which is reasonable for offline analysis (e.g. applying a fast Fourier transform), but superfluous for the methodology proposed in this paper. To achieve a feasible algorithm runtime, the data was reduced to 1000 samples by applying linear downsampling. Secondly, the repeatedly performed visual inspection implied a regular shutdown and start-up of the radial fan. As a sensor,



Fig. 6. The healthy impeller before the run-to-failure experiment.

monitoring rotational speed, was attached, these sections of the dataset could be easily identified and were removed. Thirdly, the variable impact of the different sensor signals in combination with the three selected machine learning algorithms was analyzed, and only the 9 most important signals were chosen for further consideration. The resulting dataset required, depending on the chosen configuration, up to one and a half hour of computation for a single run.

In Fig. 8 the estimations from a regression model, trained on a data set, representing a stable healthy system state, are compared with the actually sensed values (see Training). Moreover, the figure depicts estimations, received when evaluating the same model on a different, wear-containing data set (see Test). One can clearly observe, the lower estimation quality for the latter case, which proofs that a machine learning regression model is able to discriminate well between *healthy* and *defective* system behavior. Very similar results are observed when mirroring training-test comparison, i.e. when modeling an defective system state and evaluating on a healthy state representing data partition.

5.4. Experiments and results

After the successful evaluation on synthetic benchmark data, the proposed methodology was also applied on the data generated from the case study. To achieve a similar quality of result, for both, building state detection models as well as time series forecasters, a grid search for suitable algorithm parameters for Random Forest and Symbolic Regression has been performed again. The search boundaries were the same as for the synthetic data experiments in Table 1. The most promising configurations of both algorithms were selected to create regression models for the subsequent experiments in which the presented algorithms for identifying concept drift have been evaluated on real-world data.

Table 5
Overview of the mounted sensors.

	Amount	Order number
Vibration	2	VTV122
Gyroscope/Accelerometer	4	BMX055
Rotational speed	1	DI602A
Temperature	2	JUMO 90.2050
Pressure	4	Ashcroft KX1
Ambient pressure	1	Fuehler AD/A
Ambient temperature/Humidity	1	KIMO CTV 210-R



Fig. 7. The formerly healthy impeller after the run-to-failure experiment.



Fig. 8. Estimations of a model which was trained on a data partition incorporating a *healthy* system state with good estimation results (see Training) and subsequently evaluated on a *defective* partition with intendedly worsened results (see Test).

5.4.1. State detector training

For the state detector training, the first 20% of the run-to-failure dataset were used, assuming that this data partition represents a healthy state. The regular visual inspections up to this point also gave no indication of premature wear. Following the previously applied procedure for training state detection models (see Section 4.2.2), the dataset was shuffled and split in two parts, training with 67% of the data and test with the remaining 33%. The results are presented in Table 6 and the Average Relative Error (ARE), Normalized Mean Square Error (NMSE) and Pearson's R^2 (PR^2) are defined as quality measurements. For validation of these results, the first 20% of raw data were downsampled to 5000 records and the previously configured algorithms were applied again, leading to very similar results. While the ARE, NMSE and R^2 are fairly stable for both LR and SR, RF tends to overfit, however, still manages to achieve the best results. In contrast to the results from the synthetic data, significantly lower correlation but better relative and absolute error of the models can be observed.

5.4.2. Forecaster training

As illustrated in Table 7, suitable forecaster models can be created. In the presented case, a time lag of 10 to 25 can be suggested. The introduction of a higher lag does not show quality improvements, but results in increased runtime for model training, as well as for the subsequent concept drift evaluation algorithms. A closer look at the resulting models reveals that the models prefer more "recent" lags over older ones, which is quite expectable for data with such highly periodic, stable characteristics.

5.4.3. Concept drift detection and prediction

The results of the performed *concept drift detection* and *concept drift prediction* tests are summarized subsequently. Both routines have been performed with Random Forest based *state detection model* and Random Forest based *time series forecasters* with a maximum time lag = 10 and a sliding window size = 100.

According to *ARE 0* in Fig. 9, it is possible to detect the concept drift in the data stream from the radial fan. Experts, working on site at the monitored radial fan, confirmed that the illustrated deterioration matches with their visual observations. Since the real deterioration progression is not known and cannot be measured yet, we consider the *ARE* of the concept drift detection model as the new reference waveform – i.e. *ARE 0* is handled like the deterioration waveform *DET* in the experiments with the synthetic data set (cf. Section 4.2.4). Due to the confirmation of domain experts, we consider the drift detection waveform as quite similar to the expected real deterioration progression. As for Fig. 4, we compute the correlation between the concept drift detection quality measure (i.e. *ARE 0*) and the remaining concept drift prediction qualities (i.e. *ARE 5 - ARE 100*), which is visualized in Fig. 10. One can observe that the drift prediction quality drops with *Horizon 10*, however remains stable at a high level afterwards - cf. the detection quality representing y-axis is scaled between 0.80 (high quality) and 1.00 (max. quality).

According to the conducted data based experiments and detailed analysis of experts on site at the stressed radial fan, concept drift detection has been successfully deployed in the scope of the case study. However, due to the lack of continuous deterioration information, the predictability of concept drifts is currently only based on an assumption and cannot be measured yet, although the performed tests already show very promising results. Further work will tackle this open issue and provide more detailed analysis for drift prediction.

Table 6
Model estimation qualities of the *state detector* for the real-world case study data set.

	ARE		NMSE		PR^2	
	Train	Test	Train	Test	Train	Test
Linear Regression (LR)	0.013	0.013	0.664	0.678	0.335	0.323
Random Forest Regression (RF)	**0.006**	**0.012**	**0.139**	**0.589**	**0.902**	**0.416**
Symbolic Regression (SR)	0.012	0.013	0.587	0.723	0.412	0.299

6. Conclusion and outlook

This work contributes to the trending concept of predictive maintenance by proposing a methodology to find error indicating patterns in condition monitoring data streams, originating from complex industrial systems. In the first part of this work we introduced two different routines for applying regression models, trained with machine learning algorithms, for concept drift detection and concept drift prediction. While the first routine represents an adaption of state of the art drift detection methodologies, which has already been successfully tested, the latter one enhances these methodologies by adding a forecasting component. Therefore, multi-variate time series regression models for each system input variable are trained and subsequently evaluated with a so-called *rolling horizon*. The successive drift detection is performed on already forecasted values, which enables drift prediction to some extent.

For proof of concept, we accomplished to test both routines on synthetic data streams successfully. However, the major experimental part was performed in the scope of a case study concerning complex industrial radial fans. The state detection itself could distinguish between a healthy and damaged state and indicate a gradual

Table 7
Model estimation quality PR^2 of input variable *forecasters* on the test partition of the real-world case study data set.

Lag	10			25			50		
	LR	RF	SR	LR	RF	SR	LR	RF	SR
Press.sig.1	0.678	0.711	0.679	0.513	0.885	**0.892**	0.541	0.825	0.822
Press.sig.2	0.838	0.777	0.833	**0.842**	0.802	0.819	0.751	0.804	0.782
Press.sig.3	**0.996**	0.993	0.995	**0.996**	0.993	0.995	0.991	0.992	0.993
Vibr.sig.1	0.991	**0.992**	0.991	0.990	0.989	0.990	0.988	0.989	0.986
Vibr.sig.2	0.968	0.957	0.968	**0.974**	0.946	0.963	0.940	0.952	0.940
Temp.sig.1	0.996	**0.997**	**0.997**	0.996	0.980	0.996	0.984	0.995	0.981
Temp.sig.2	0.993	**0.994**	**0.994**	0.991	0.973	0.993	0.975	0.991	0.977
Temp.sig.3	0.861	0.914	0.896	0.653	0.865	**0.921**	0.587	0.908	0.894
Vol.flow	0.907	0.888	0.908	**0.913**	0.905	0.905	0.902	0.909	0.888

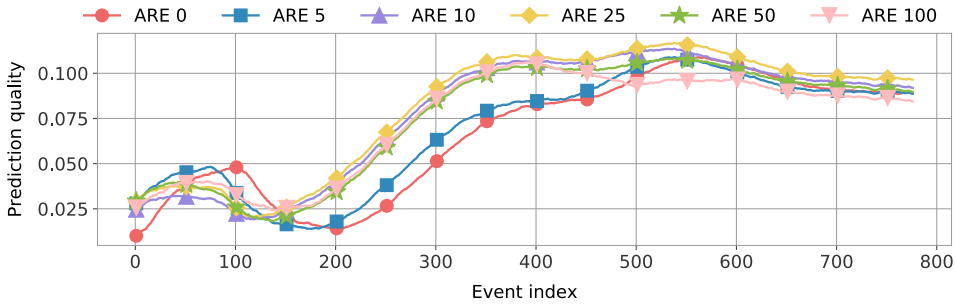


Fig. 9. Average Relative Errors (ARE) of the state detector model evaluated on streams, which were forecasted with increasing horizon (cf. ARE *horizon*); *horizon* = 0 refers to concept drift detection; *horizon* > 0 refers to concept drift predictions.

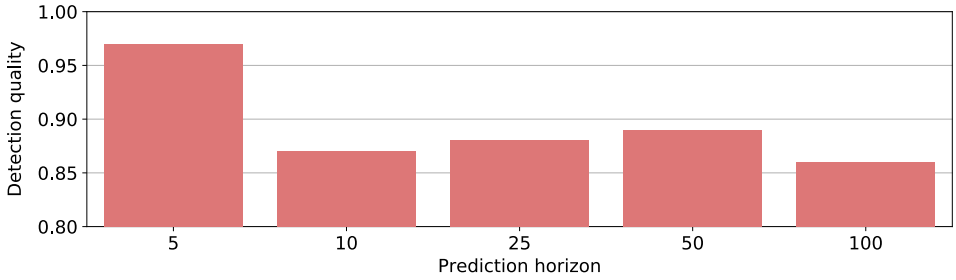


Fig. 10. Detection quality vs. prediction qualities, i.e. the correlation of the *ARE 0* waveform against all the other displayed horizons (i.e. horizon 5, 10, 25, 50, 100) from Fig. 9.

deterioration. A benefit of the drift prediction can be observed by choosing a sufficiently large forecast horizon, predicting the drift early on. An increase of the forecast horizon, however, will also cumulate small errors introduced by each forecasting-step and therefore increases the absolute error between actual values and forecasted values. By improving the forecasters, either the forecast horizon can be extended or the error can be reduced. However, until now, drift prediction could not be verified for the case study, since the necessary information concerning deterioration progression is not measured yet.

Based on these promising algorithm assessments, we plan larger scaled experiment series with a broader comparison of algorithms and configurations. This might accelerate decisions for other researchers and practitioners on one hand and possibly reveals additional potential of the presented routines on the other. For future work we also consider tackling the following open issues.

- As an **alternative time series preprocessing technique** decomposition techniques could be utilized. Additional information regarding season, trend, as well as new features from the frequency domain may help to narrow the focus of drift detection.
- Employing **ensemble models** might make training and prediction results more robust and enable confidence-based evaluations.
- Due to the requirement of real-time evaluation ability for any analysis method applied to the area of predictive maintenance, **performance tuning** of the developed routines might be another worthwhile step.
- Additional runs-to-failure would provide a more diverse dataset for the training of the algorithm. This could further improve the results or at least reduce a possible bias and increase the generalizability.
- Another topic of interest is the idea of **open ended learning**, which blurs the line between offline and online phase in order to slowly adapt models to concept drifts. The pursued detection of changes in a system's dynamics could be performed based on the model genotype, i.e. by analyzing changing model structures, instead of analyzing changing model estimation qualities.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

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